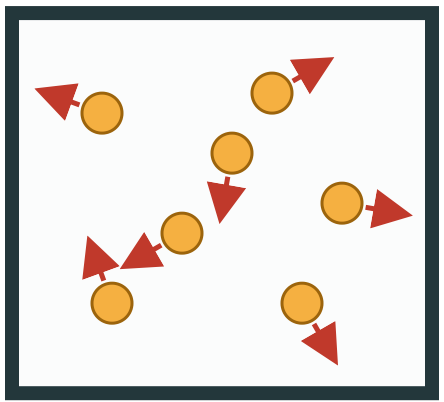
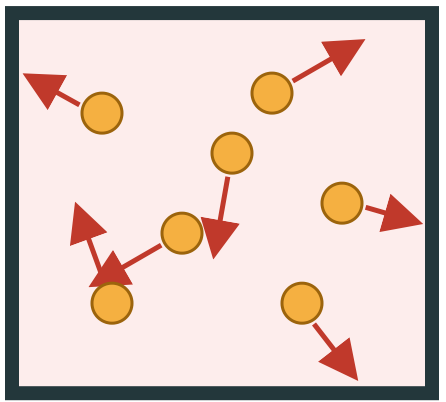
Gay-Lussac's law: the volume does not change



we heat it
T × 2



T = 300 K
p = 1 atm

T = 600 K
p = 2 atm

V constant (closed, rigid container)

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Example: $\frac{1\ \text{atm}}{300\ \text{K}} = \frac{2\ \text{atm}}{600\ \text{K}}$

After a long drive the air in the tyres is hotter, so its **pressure is higher**.

Never heat a spray can: the pressure rises and it can **explode**.